

Securing Small Wireless Networks -- Guided Notes

ANSWER KEY

Unit tp-6.15 | CompTIA Network+ N10-009 Obj. FC0-U71 6.5 | N10-008 FC0-U71 6.5

For instructor use / distribute after students complete the notes.

Question 1

Wi-Fi networks use _____ waves to transmit data, making them accessible to anyone within range -- including attackers. Unlike wired networks, there is no _____ connection requirement. Therefore, wireless networks require strong encryption and authentication to prevent _____ access.

Answer: Radio; physical; unauthorized.

Question 2

WEP (Wired Equivalent Privacy) was the original Wi-Fi encryption standard and is considered _____ -- do not use it. WPA (Wi-Fi Protected Access) improved on WEP but also has known weaknesses. WPA2 uses _____ encryption and is considered _____ secure. WPA3 is the _____ standard with stronger protections against offline dictionary attacks.

Answer: Broken; AES; adequately; latest (current).

Question 3

The Wi-Fi password on a home or small business router uses the _____ (Pre-Shared Key) mode of WPA2/WPA3. The PSK should be: at least _____ characters, random (not a dictionary word), and changed from the _____ password if one was preset. Enterprise networks use WPA2/WPA3 _____ mode with a RADIUS server instead of a shared key.

Answer: PSK; 12+; default; Enterprise.

Question 4

The router admin console should be secured by: changing the _____ admin username and password, disabling _____ management (only allow admin access from the local network, not from the internet), and updating the router _____ regularly to patch known vulnerabilities.

Answer: Default; remote; firmware.

Question 5

SSID _____ hides the network name so it does not appear in Wi-Fi scans. This provides _____ through obscurity -- not true security -- because attackers with basic tools can still detect hidden networks by monitoring _____ traffic. MAC address filtering only allows _____ MAC addresses to connect, but MAC addresses can be _____ by attackers.

Answer: Hiding; security; wireless; whitelisted; spoofed.

Question 6

A guest network is a _____ Wi-Fi network that allows visitors to access the internet without accessing the _____ network where company or family devices reside. IoT devices (smart TVs,

thermostats, cameras) should also be placed on an _____ VLAN or guest network because they often have poor _____ update practices and can serve as pivot points.

Answer: Separate (isolated); primary; IoT; firmware.

Question 7

An evil twin attack creates a rogue wireless _____ with the same SSID as a legitimate network. Victims connect to the attacker's access point instead of the real one, allowing _____ -in-the-middle interception of all traffic. Using a _____ on any untrusted network prevents attackers from reading intercepted traffic even if connected to a rogue AP.

Answer: Access point; man; VPN.

Question 8 -- Real World Application

REAL WORLD: A small dental office has one wireless router for staff and patients combined. The router uses WPA2 with the password 'Dentist2018' and has never had a firmware update. A patient connects their laptop and notices the router admin page is accessible. They log in with the default admin/admin credentials. Identify four specific security failures and describe the fix for each.

Answer: Failure 1 -- Shared network for staff and patients: patients should never be on the same network segment as staff computers and the practice management system containing patient PHI. Fix: set up a separate guest SSID for patient Wi-Fi that is isolated from the internal network. Failure 2 -- Weak, predictable password ('Dentist2018'): a dictionary-based password with a year suffix is trivially guessable. Fix: generate a random 16+ character WPA2-PSK and store it in a password manager. Failure 3 -- Default admin credentials (admin/admin): unchanged default credentials are the first thing any attacker tries. Fix: change the admin username and password to a strong unique credential immediately on setup. Disable remote admin access so the console is only reachable from the local wired network. Failure 4 -- No firmware updates: unpatched router firmware contains known vulnerabilities, including some that allow unauthorized remote code execution. Fix: check the manufacturer website or router admin panel for firmware updates and apply them. Enable automatic firmware updates if available.